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Broadband Trilayer Adiabatic Edge Coupler on Thin-Film Lithium Tantalate for NIR Light

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Abstract

This work addresses the challenge of realizing broadband, low-loss fiber-to-waveguide coupling in the short-wavelength near-infrared range (700–1050 nm), where the required fine structural dimensions and taper tips approach or even exceed current fabrication limits, resulting in tight fabrication tolerances and degraded coupling efficiency. We propose a broadband trilayer adiabatic edge coupler on a thin-film lithium tantalate platform that requires only two standard lithography and etching steps. The design integrates a crossed bilayer taper and a dual-core mode converter to achieve adiabatic mode transformation from a ridge to a thin strip waveguide, ensuring excellent fabrication tolerance and process simplicity. Simulations predict a minimum coupling loss of 0.57 dB at 850 nm, which includes the transmission through the complete edge-coupler structure, along with a 0.5-dB bandwidth exceeding 140 nm. The proposed structure provides a broadband, low-loss, and fabrication-tolerant interface for short-wavelength photonic systems such as quantum photonics, biosensing, and visible-light communications.

Keywords: thin-film lithium tantalate; edge coupler; adiabatic taper; near-infrared light; thin-film lithium niobate

1. Introduction

The visible-to-near-infrared wavelength band is crucial for advanced applications such as quantum optics, biosensing, and visible-light communications [1–3]. Transparent material platforms including silicon nitride (Si_3N_4), aluminum oxide (Al_2O_3), and niobium pentoxide (Nb_2O_5) have been extensively explored, enabling significant progress in low-loss waveguides and integrated devices [3]. However, their electro-optic modulation and nonlinear performance remain limited. Recently, thin-film lithium niobate (TFLN) and thin-film lithium tantalate (TFLT) have emerged as promising alternatives, offering a wide transparency window, large electro-optic coefficients, and strong second-order nonlinearities [4,5]. In particular, TFLT exhibits a higher optical damage threshold [6], reduced DC drift [7,8], and a lower photorefractive effect [9], making it especially attractive for short-wavelength applications. More detailed material properties of LT and LN can be found in the Extended Data Table of Ref. [10].

Efficient fiber-to-chip coupling is critical to overall system performance. However, achieving broadband and low loss coupling at short wavelengths remains highly chal-

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lenging: as the wavelength decreases, the required grating feature sizes and taper tip dimensions approach or even exceed current fabrication limits. Although TFLN-based devices have demonstrated edge coupling with a −3 dB loss per facet at 775 nm [11] and grating coupling of −3.45 dB at 532 nm [12], their performance still lags behind the state-of-the-art values below 1 dB at 1550 nm [13,14]. Moreover, fabrication tolerance and operational bandwidth issues become more pronounced at shorter wavelengths.

To address these challenges, we propose a novel trilayer broadband edge coupler on TFLT, fabricated using only two standard lithography steps. The design integrates a crossed bilayer taper with an asymmetric dual-core mode converter, enabling efficient mode transformation and reduced mismatch to lensed fibers. Simulated results show a minimum coupling loss of 0.57 dB at 850 nm, a 0.5 dB bandwidth exceeding 140 nm, and robust tolerance to variations in width, thickness, and alignment—providing a low-loss, broadband solution for short-wavelength photonic integration.

2. Simulation Methods

The coupling efficiency of an edge coupler is mainly determined by the facet mode mismatch between the input fiber and the on-chip waveguide, as well as the transmission loss during the mode-size conversion process.

The finite-difference eigenmode (FDE) method is used to calculate the transverse modal field distributions of the fiber and the chip facet. The fiber–chip facet coupling efficiency is then obtained from the mode overlap integral between the two calculated modes.

The transmission characteristics of the mode conversion structures are simulated using the eigenmode expansion (EME) method. The transmission efficiency is obtained from the scattering parameter S_{21} as $\eta = |S_{21}|^2$, and the corresponding transmission loss is calculated as $10\log_{10}(\eta)$.

The wavelength-dependent refractive indices of LT used in the simulations are obtained from the refractiveindex.info database.

3. Design and Structure

The proposed edge coupler is designed for the 700–1050 nm wavelength range on the TFLT platform. Other functional devices on the same platform typically employ ridge waveguides rather than fully etched ones, as ridge waveguides offer reduced propagation loss, improved fabrication yield and repeatability, and enhanced electro-optic modulation efficiency. However, ridge waveguides confine the optical mode tightly, resulting in a small mode size that limits efficient edge coupling to optical fibers. To mitigate this mismatch, the waveguide is fully etched and its thickness is reduced near the chip facet to enlarge the mode size, while maintaining a sufficiently large feature size and fabrication tolerance. Figure 1a shows the optimal taper tip width as a function of waveguide thickness, corresponding to simulated coupling losses ranging from 0.55 to 0.83 dB at a wavelength of 850 nm, when coupled to a lensed fiber with a 2.5-μm mode field diameter. The required taper tip width increases as the waveguide thickness decreases, thereby relaxing the fabrication tolerance. Considering a fabrication tolerance better than ±100 nm and the achievable etching accuracy in thickness, a 50-nm thickness is selected for the edge waveguide.

The key challenge then lies in efficiently transferring the optical mode from the ridge waveguide to the 50-nm-thick strip waveguide. Figure 1b illustrates the conventional bilayer inverse-taper scheme, in which the optical field is adiabatically converted from a 150-nm ridge to a 50-nm strip waveguide. In this design, the transmission efficiency strongly depends on the tip width W_0 of the top-layer taper. Simulations indicate that, under an 180-nm process node, the coupling loss can reach up to 1.8 dB. Although adding more

taper layers could further reduce the loss, this approach inevitably increases fabrication complexity by requiring additional lithography and etching steps.

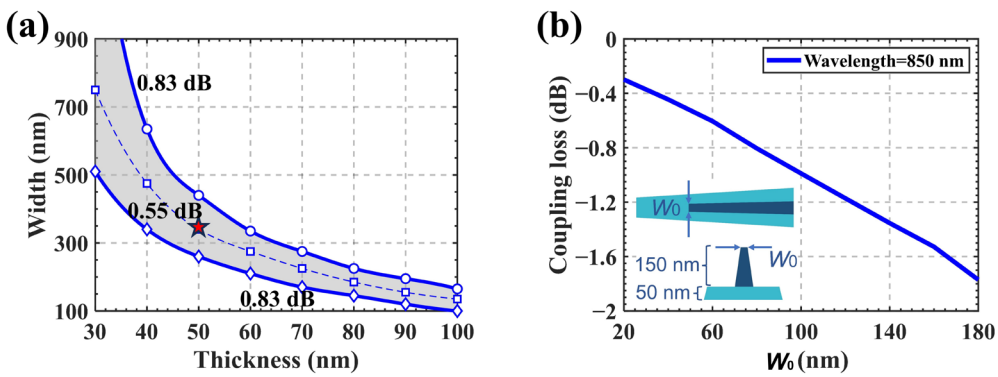

Figure 1. (**a**) Optimized taper tip thickness and width. The dashed line indicates the dimensions for a minimum coupling loss of 0.55 dB, and the solid line for 0.83 dB. The shaded area represents the tolerance range, and the star marker shows the design parameters. (**b**) Simulated transmission loss of the conventional ridge-to-strip waveguide versus taper tip width. The inset shows the schematic structure.

To balance coupling efficiency and fabrication simplicity, we propose a trilayer transition structure that requires only two standard lithography and etching steps. A crossed bilayer taper is employed to minimize taper-tip-introduced loss. As illustrated in Figure 2a, the ridge waveguide (waveguide C, with a 100-nm ridge height and a 100-nm-thick slab) gradually couples into a 100-nm fully etched waveguide (waveguide B). Unlike conventional top-layer tapers, the proposed structure features an obliquely extended taper tip that crosses the second-layer waveguide. During the second etching step, the top-layer taper is redefined, forming a sharp three-dimensional intersection tip. This strategy, previously validated at the C-band [13], achieves a near-adiabatic transition from ridge to strip waveguides, effectively overcoming the fabrication-limited taper-tip constraint.

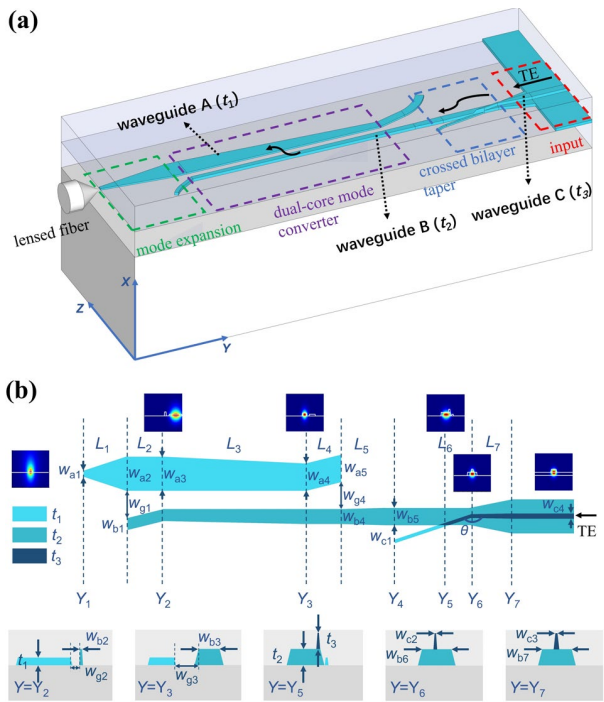

Figure 2. (**a**) Three-dimensional schematic and (**b**) top view of the proposed trilayer adiabatic edge coupler. The insets in (**b**) show the simulated mode profiles at key cross sections.

Subsequently, a dual-core mode converter is introduced to transfer the optical mode from the 100-nm waveguide (B) to the 50-nm waveguide (A). The conversion principle relies on adiabatic mode evolution. At the Y_3 cross-section, as shown in Figure 2b, the effective index satisfies $n_{\mathrm{eff}}(B) > n_{\mathrm{eff}}(A)$, and the optical energy is mainly confined in waveguide B. Along the propagation, waveguide B is gradually narrowed while waveguide A is widened, until at the Y_2 cross-section $n_{\mathrm{eff}}(B) < n_{\mathrm{eff}}(A)$, leading to a smooth transfer of energy into waveguide A. Finally, a single inverse taper in waveguide A expands the mode size to match the lensed fiber. The mode profiles at key cross sections are shown in Figure 2b. The structural parameter values of different areas marked in the plan view are listed in Table 1.

Table 1. Key geometrical parameters of the proposed coupler.

Parameters	t_1	t_2	t_3	θ	W_{a1}	W_{a2}, W_{a3}	W_{a4}, W_{a5}	W_{b1}, W_{b2}	W_{b3}, W_{b4}	W_{b5}	W_{b6}, W_{c4}	W_{b7}	W_{c1}, W_{c2}	W_{c3}	W_{g2}, W_{g3}	W_{g1}, W_{g4}
Value (μm)	0.05	0.1	0.1	179.24°	0.34	1.5	0.6	0.15	0.45	0.5	1	5	0.15	0.6	0.35	1.35

4. Simulation Results

4.1. Edge Coupling Performance

The optimal waveguide width is 340 nm for a 50-nm-thick edge tip, corresponding to a minimum fiber–chip coupling loss of 0.55 dB at 850 nm, as estimated from the mode overlap integral. Figure 3a shows the fiber alignment tolerances Δx and Δz along the x- and z-axes, respectively. When the misalignment reaches 0.6 μm or 1 μm, the coupling loss increases to approximately 1.5 dB and 3 dB, respectively. The alignment tolerance is primarily determined by the fiber's mode-field size, and in practice, a positioning accuracy of 0.6 μm is readily achievable with commercial alignment systems.

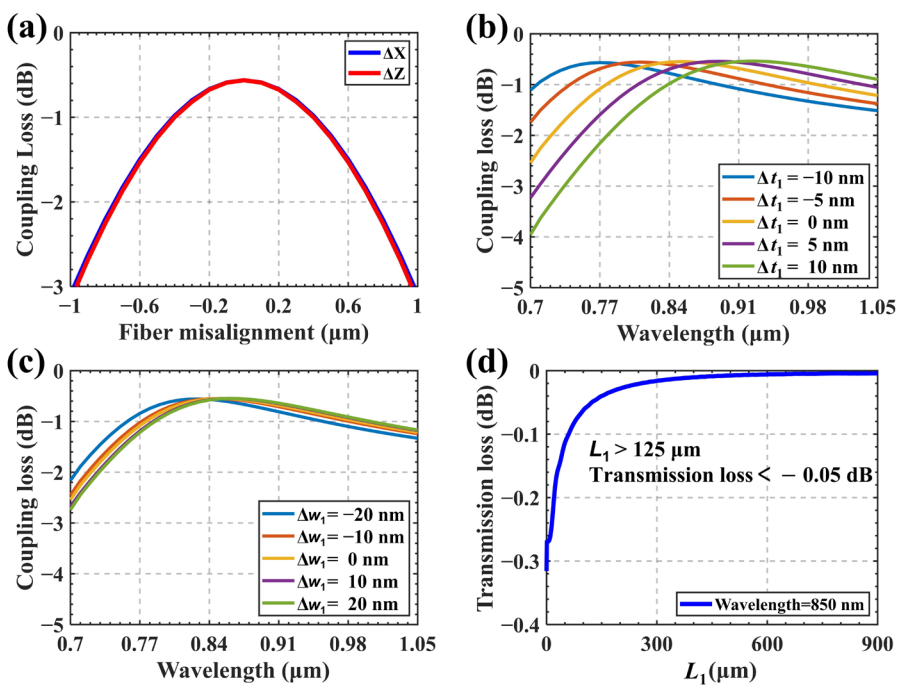

Figure 3. (**a**) Coupling loss as a function of fiber misalignment at 850 nm. (**b**) Wavelength response for thickness variation. (**c**) Wavelength response for width variation. (**d**) Transmission loss as a function of taper length (L_1).

Figure 3b shows the wavelength response under thickness variations (Δt_1) of up to ±10 nm. The central wavelength exhibits a slight shift, while the minimum coupling loss

remains nearly constant. The 0.5-dB bandwidth exceeds 140 nm when Δt_1 = 0 nm. Notably, at shorter wavelengths (<850 nm), a thinner edge waveguide improves coupling efficiency, while at longer wavelengths (>850 nm), the performance degradation is minimal.

As shown in Figure 3c, a ±20 nm variation (Δw_1) in the taper tip width (W_{c1} and W_{c2}) has a negligible effect on both the coupling loss (<0.1 dB degradation at 850 nm) and the bandwidth. This demonstrates that the proposed design exhibits excellent tolerance to lithographic and etching variations, ensuring robust and reproducible fabrication performance.

For the mode expansion section, the taper width of waveguide A increases linearly from W_{a1} to W_{a2} over a length L_1. The simulated transmission loss as a function of L_1 is shown in Figure 3d. When L_1 = 125 µm, the corresponding loss is less than 0.05 dB.

4.2. Dual-Core Mode Converter

Simulations of the dual-core mode converter are performed to analyze the optical transition from waveguide A to waveguide B and to evaluate the tolerance to fabrication variations.

The effects of lithographic alignment error (Δg) between waveguides A and B, as well as width (Δw_2) and thickness (Δt_1) variations in waveguide A, are investigated. Figure 4a shows that efficient and stable mode conversion at 850 nm is achieved when the coupling length $L_3 \geq 165$ µm. At the optimized coupling length of L_3 = 800 µm, the converter maintains stable performance across 800–1050 nm under alignment errors $|\Delta g| < 100$ nm (Figure 4b). Furthermore, it exhibits high tolerance to both Δt_1 and Δw_2 (Figure 4c,d), confirming its compatibility with standard fabrication processes and robustness against dimensional deviations.

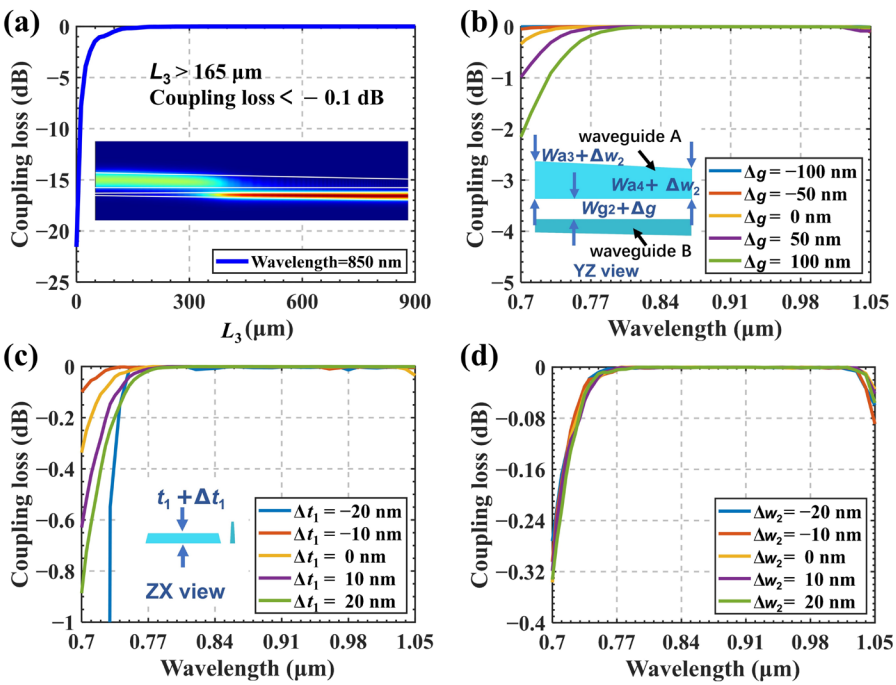

Figure 4. (**a**) Coupling loss of the dual-core mode converter as a function of taper length (L_3). The inset shows the optical field evolution. Wavelength responses for (**b**) gap variation between waveguides A and B, (**c**) thickness variation, and (**d**) width variation in waveguide A. The insets in (**b**,**c**) show schematic illustrations of the tolerance analysis.

At the input and output of the dual-core mode converter, two gap-tapered sections are introduced to adiabatically expand the gap from W_{g2} (W_{g3}) to W_{g1} (W_{g4}), with lengths of $L_2 = L_4$ = 200 µm, thereby mitigating abrupt mode-field transitions.

4.3. Crossed Bilayer Taper

The crossed bilayer taper represents one of the key advantages of our design, achieving near-ideal adiabatic optical transmission while maintaining remarkable robustness against fabrication variations. This structure exhibits an ultra-low transmission loss of <0.001 dB across the near-infrared band. As shown in Figure 5, the crossed bilayer taper exhibits excellent tolerance to lithographic alignment error (Δa), as well as to thickness (Δt_3) and width (Δw_3) variations in the top taper, at the optimized total coupling length of L_6 = 200 µm. Simulation results reveal negligible performance degradation for Δa up to 100 nm and Δt_3 or Δw_3 within ±20 nm. These relaxed tolerances substantially mitigate the reliance on nanoscale patterning precision.

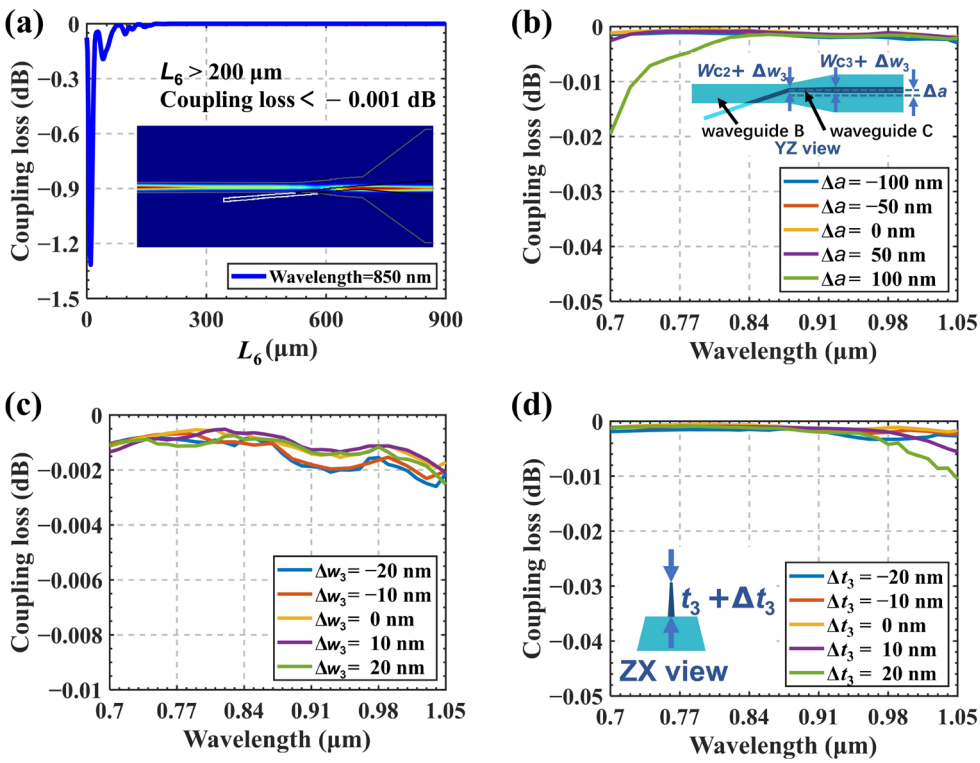

Figure 5. (**a**) Coupling loss as a function of the crossed bilayer taper length (L_6). The inset shows the optical field evolution at 850 nm and a schematic illustration of the tolerance analysis for the crossed bilayer taper. (**b**) Lithographic alignment error, (**c**) width variation, and (**d**) thickness variation in the top taper.

The linearly width-tapered sections on both sides of the crossed bilayer taper, with lengths of L_5 = 5 µm and L_7 = 50 µm, respectively, are employed to ensure adiabatic coupling between the waveguides.

5. Fabrication Process Design

The proposed trilayer edge coupler can be fabricated on an LT on insulator (LTOI) platform with a 200-nm LT film and a 3-µm buried oxide layer, requiring only two standard lithography and etching steps. The process flow is schematically illustrated in Figure 6.

First, a photoresist layer is spin-coated onto the wafer, followed by the first exposure and development to define the patterns for waveguide A and waveguide C. The LT film is then etched by 100 nm to form the ridge height of waveguide C and the upper section of waveguide A.

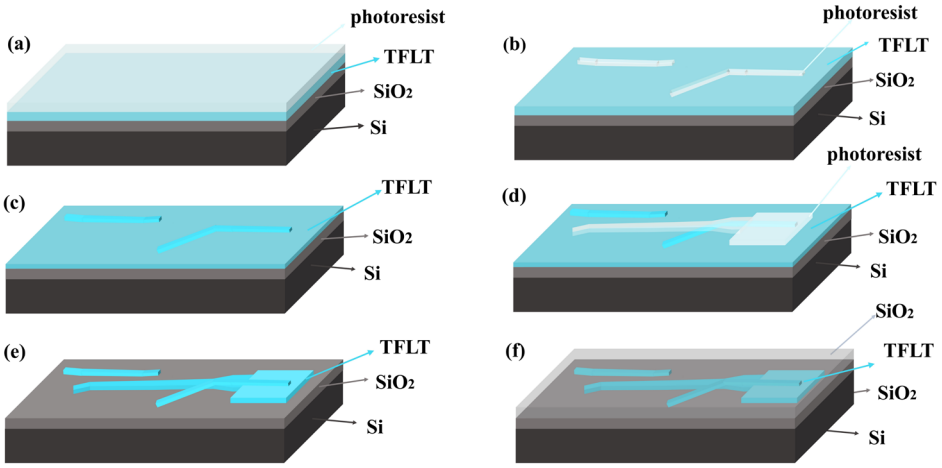

Figure 6. Schematic of the fabrication process flow: (**a**) Photoresist coating, (**b**) Patterning of waveguides A and C, (**c**) First etching (100 nm), (**d**) Patterning of waveguide B and the slab layer, (**e**) Second etching (150 nm), (**f**) Cladding deposition.

Subsequently, a second lithography step defines waveguide B and the slab of input region. During this exposure, most of waveguide C is protected by photoresist, except for the part of the tilted tip that overlaps with the newly patterned region. A second etching of 150 nm is then performed. As the remaining slab layer is 100 nm thick, these regions are slightly over-etched during this step, whereas waveguide A is further thinned, resulting in a 50-nm-thick strip waveguide.

Finally, a SiO_2 cladding layer is deposited to protect the device, followed by conventional dicing, polishing, and facet preparation. This streamlined process enables the realization of a three-layer edge-coupler structure using only two lithography and etching steps, ensuring high fabrication efficiency and strong compatibility with standard CMOS processes.

6. Conclusions and Discussion

A broadband trilayer adiabatic edge coupler on the TFLT platform has been proposed and numerically demonstrated for efficient fiber-to-chip coupling in the near-infrared region. By integrating a crossed bilayer taper with an adiabatic dual-core mode converter, the design achieves smooth mode transition from a ridge to a 50-nm strip waveguide using only two standard lithography and etching steps. Simulations predict a minimum coupling loss of 0.57 dB at 850 nm for the complete edge-coupler structure, together with a 0.5-dB bandwidth exceeding 140 nm and excellent tolerance to dimensional and alignment variations.

As summarized in Table 2, the proposed coupler demonstrates superior broadband performance compared with previously reported grating and edge couplers implemented on TFLN and Si_3N_4 platforms, which typically suffer from coupling losses of 2–10 dB and bandwidths below 30 nm in the visible to near-infrared regime. This performance improvement primarily originates from the near-adiabatic energy transfer enabled by the trilayer architecture, which effectively suppresses radiation loss during mode expansion while simultaneously relaxing fabrication constraints.

Table 2. Device performance comparison.

Coupler Type	Wavelength (nm)	Bandwidth (nm)	Coupling Loss (dB)	Fiber Type/MFD
TFLN # [11]	775	\	1	LF
TFLN * [12]	532	~20 (3-dB)	2.5	SMF
Si_3N_4 # [15]	640	~100 (0.5-dB)	0.5	SMF
Si_3N_4 * [16]	637	~20 (3-dB)	3	4.5 μm
TFLN * [17]	775	~11 (3-dB)	3	5.5 μm
Si_3N_4 * [18]	888	~30 (1-dB)	5.9	SMF
This work #	850	140 (0.5-dB)	0.57	LF

*: grating coupler; #: edge coupler; SMF: single mode fiber; LF: lensed fiber; MFD: mode field diameter.

The total length of the edge-coupler device is 1.58 mm, which is slightly longer than that of conventional edge couplers. This is mainly because three adiabatic mode-conversion stages are employed to achieve a low coupling loss. By adopting conversion-enhanced adiabatic tapers [19] or segmented shape optimization inverse designs [20], the overall device length could be further reduced by more than half.

Although this work focuses on the fundamental TE mode, the proposed adiabatic trilayer coupling strategy can also be extended to TM polarization. With appropriate re-optimization of the waveguide parameters, a polarization-insensitive edge coupler can be realized, which is promising for applications in quantum optics and biosensing.

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Conflicts of Interest: The authors declare no conflicts of interest.

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